

N. Shree Deepthi Batch – 37

AI-ASSISTED CODING

Assignment 19

Task 1 – Python to C++ Conversion

Task:

Translate a simple Python class into C++ using AI assistance.

Instructions:

- Prompt AI to convert a Python class representing a Student with attributes and methods into equivalent C++ code
- Ensure proper handling of:
 - o Constructors
 - o Data types
 - o Access specifiers
- Compile and run both versions to verify output consistency

Expected Output:

An equivalent working C++ class translated from Python.

Python code:

```
1  class Student:
2      def __init__(self, name, age):
3          self.name = name
4          self.age = age
5
6      def display(self):
7          print(self.name, self.age)
8
9  s = Student("Tom", 20)
10 s.display()
```

Prompt used:

Convert the following Python class into equivalent C++ code.

Python Code:

class Student:

def __init__(self, name, age):

self.name = name

self.age = age

def display(self):

print(self.name, self.age)

Requirements:

- Use proper C++ syntax
- Include constructor
- Use correct data types
- Use public access specifier
- Include main() function

Also ensure the output is same as Python.

Output:

```
assignment 19.3.cpp > Student > Student(string, int)
1  #include <iostream>
2  using namespace std;
3  class Student {
4  public:
5      string name;
6      int age;
7
8      Student(string n, int a) {
9          name = n;
10         age = a;
11     }
12
13     void display() {
14         cout << name << " " << age;
15     }
16 };
17
18 int main() {
19     Student s("Shruthi", 20);
20     s.display();
21 }
```

```
Tom 20
PS C:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding>
```

Task 2 – Java to Python Function Conversion

Task:

Convert a Java method that checks whether a number is prime into Python.

Instructions:

- Ask AI to translate a Java isPrime() method into Python
- Test the function with multiple inputs
- Ensure the Python version follows Pythonic syntax and logic

Expected Output:

A correct Python function equivalent to the original Java prime-checking method.

```
1  boolean isPrime(int n) {  
2      if (n <= 1) return false;  
3      for (int i = 2; i < n; i++) {  
4          if (n % i == 0) return false;  
5      }  
6      return true;  
7  }
```

Prompt used:

Convert the following Java method into Python.

Java Code:

```
boolean isPrime(int n) {  
    if (n <= 1) return false;  
    for (int i = 2; i < n; i++) {  
        if (n % i == 0) return false;  
    }  
    return true;  
}
```

Requirements:

- Follow Python syntax

- Use indentation properly
- Return True/False correctly
- Add a sample test case

Output:

```
1  def isPrime(n):
2      if n <= 1:
3          return False
4      for i in range(2, n):
5          if n % i == 0:
6              return False
7      return True
8
9  print(isPrime(7))
```

```
PS C:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding> & "c:\Users\Lenovo\OneDrive\Desktop\AI
venv\Scripts\python.exe" "c:/Users/Lenovo/OneDrive/Desktop/AI Assisted Coding/Assignment_18.3.
True
```

Task 3 – Pseudocode to Python Implementation

Task:

Translate a given pseudocode algorithm into Python using AI.

Instructions:

- Provide AI with pseudocode for sorting numbers (e.g., bubble sort or selection sort)
- Ask AI to convert it into executable Python code
- Validate correctness using sample input lists

Expected Output:

A working Python program that correctly implements the pseudocode

Convert the following pseudocode into Python code.

Pseudocode:

1. Start
2. Take a list of numbers
3. Compare adjacent elements
4. Swap if needed
5. Repeat until sorted
6. Print sorted list

Requirements:

- Use simple Python loops
- Keep code easy to understand
- Print final sorted output

```
1 arr = [5, 2, 8, 1]
2
3 for i in range(len(arr)):
4     for j in range(len(arr)-1):
5         if arr[j] > arr[j+1]:
6             arr[j], arr[j+1] = arr[j+1], arr[j]
7
8 print(arr)
```

Output:

```
[1, 2, 5, 8]
PS C:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding>
```

Task 4 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

Instructions:

- Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).
- Ask AI to generate equivalent Pandas code.
- Test the generated code on a sample DataFrame.

Expected Output:

- Equivalent Pandas query that retrieves the correct subset of data.

Prompt used:

Convert the following SQL query into Pandas Python code.

SQL Query:

SELECT name, salary FROM employees WHERE salary > 50000;

Requirements:

- Create a sample DataFrame
- Use Pandas filtering
- Display the result
- Keep code simple

```
1  import pandas as pd
2
3  data = {
4      "name": ["A", "B", "C"],
5      "salary": [40000, 60000, 70000]
6  }
7
8  df = pd.DataFrame(data)
9
10 result = df[df["salary"] > 50000][["name", "salary"]]
11
12 print(result)
```

```
PS C:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding> & "c:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding\venv\Scripts\python.exe" "c:/Users/Lenovo/OneDrive/Desktop/AI Assisted Coding/Assignment 19.3.py"
name salary
B      60000
C      70000
PS C:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding> |
```

Task 5 – Real-Time Application: Algorithm Translation Across Languages

Scenario:

A company maintains algorithms written in different programming languages.

Examples:

- Converting a Python searching algorithm into C
- Translating a Java sorting program into JavaScript

Instructions:

- Use AI to translate a selected algorithm between two programming languages
- Execute and test both versions
- Document translation challenges such as:
 - o Syntax differences
 - o Library support
 - o Memory management

Expected Output:

Equivalent, tested code implementations in two different programming languages.

Prompt used:

Convert the following Python code into C language.

Python Code:


```
arr = [1,2,3,4,5]
```

```
key = 3
```

```
for i in arr:
```

```
    if i == key:
```

```
        print("Found")
```

Requirements:

- Use C syntax
- Use array properly
- Include main function
- Use printf for output

```
arr = [1,2,3,4,5]

key = 3
for i in arr:
    if i == key:
        print("Found")
```

```
1  #include <stdio.h>
2
3  int main() {
4      int arr[5] = {1,2,3,4,5};
5      int key = 3;
6
7      for(int i=0;i<5;i++){
8          if(arr[i]==key){
9              printf("Found");
10         }
11     }
12 }
```

```
venv\Scripts\python.exe" "c:/Users/Lenovo/OneDrive/Desktop/AI Assiste
Found
PS C:\Users\Lenovo\OneDrive\Desktop\AI Assisted Coding> |
```